



## Shocking story

Scientists in the 19th and early 20th centuries learned that an **electric shock** could restart a heart or correct it when it started beating abnormally. The first defibrillators passed an electric current through **wires touching the heart** and could be used only in hospitals during an **OPERATION**.

Paddles are placed on the patient's body.

## Closed chest

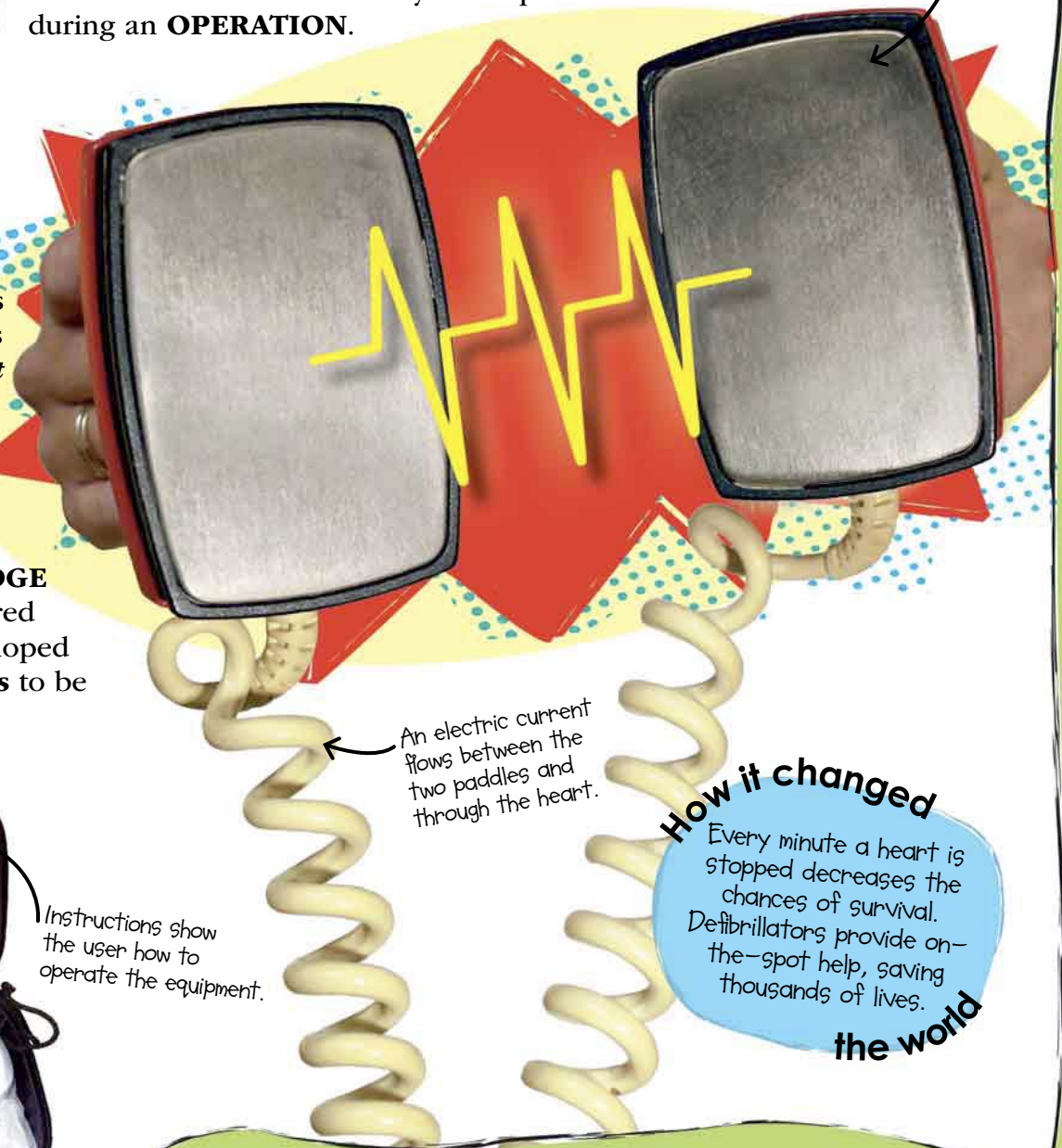
Closed-chest defibrillators can correct heart rhythms or restart a heart **without the body being opened up**. The first units were huge, expensive, and needed to be plugged in. In 1965, Northern Irish doctor **FRANK PANTRIDGE** built a defibrillator powered by car batteries and developed **smaller, portable models** to be carried in ambulances.



Instructions show the user how to operate the equipment.

## Modern design

Modern defibrillators are **lightweight** and found in many places, from swimming pools to schools, not just in hospitals. Some give **spoken instructions** so that any adult can use them in an emergency.



An electric current flows between the two paddles and through the heart.

**How it changed**  
Every minute a heart is stopped decreases the chances of survival. Defibrillators provide on-the-spot help, saving thousands of lives.  
**the world**

# Defibrillator

The **SHOCKING** invention that saves lives by restarting hearts